

or crop sensors. A full-sensor is the kind of image sensor that is equivalent to a typical 35 mm frame (36x24mm) of photographic film. Virtually all P&S cameras, including some D-SLRs use a frame that is smaller than the 35mm one because its smaller, cheaper to make and leads to cheaper, lighter cameras.

Cameras – like Canon’s DSLRs, which use EOS – 1Ds or EOS-5D full frame sensors (where EOS stands for Electro Optical System) – end up becoming heavier and bulkier because you need better, bigger, hi-tech lenses and a body to match. Point and shoot cameras usually use smaller sensors, typically of a size equivalent to APS-C-size film – significantly smaller than a full 35 mm frame. Smaller sensors record only a part of the image, giving a ‘zoomed in’ effect



Size does matter

‘Full-frame’ also refers to the CCD sensors in which the sensor elements occupy the entire face of the sensor, instead of devoting half the space to storage sites.

A camera is built to suit the needs of the sensor, so a full-frame sensor would need a lens that successfully projects the image on the entire sensor. If the full-frame lens is set on a camera with a small sensor, it just captures the central portion of the image, cropping off the edges. The ratio of the size of the image to the size of the sensor is called “crop factor”, where the reference is usually the 35mm full frame sensor.